(TSM) for enhanced security and observability across Kubernetes clusters, connecting modern applications with traditional components.

Multi-cluster management

As Kubernetes enables running containerised apps anywhere, multi-cloud and multi-cluster configurations (MCMCK) are becoming essential. However, managing Kubernetes across various environments is challenging. Let's explore MCMCK and strategies for mastering it through purpose-built management solutions.

MCMCK refers to Kubernetes deployments spanning multiple public clouds, private data centres, and edge locations. Enterprises adopt multi-cloud to avoid vendor lock-in, optimise costs, ensure disaster recovery, and meet data sovereignty requirements.

Challenges of managing multi-cluster Kubernetes: Managing multiple Kubernetes clusters across diverse environments leads to issues like cluster sprawl, lack of centralised control, networking complexities, and data movement challenges. Dedicated MCMCK platforms help overcome these hurdles by offering unified control, app deployment across clusters, data replication, and seamless communication.

Key use cases for MCMCK: MCMCK enables cost optimisation by deploying apps to the most suitable cloud, avoids vendor lock-in, ensures high availability through cross-cloud replication, and scales infrastructure globally for increased reach and performance.

Leading MCMCK management solutions include:

- **Google Anthos**: Offers managed Kubernetes across on-prem, Google Cloud, AWS, and Azure with centralised control.
- **Red Hat OpenShift**: Provides enterprise Kubernetes with multi-cluster management, enhanced security, and GitOps.

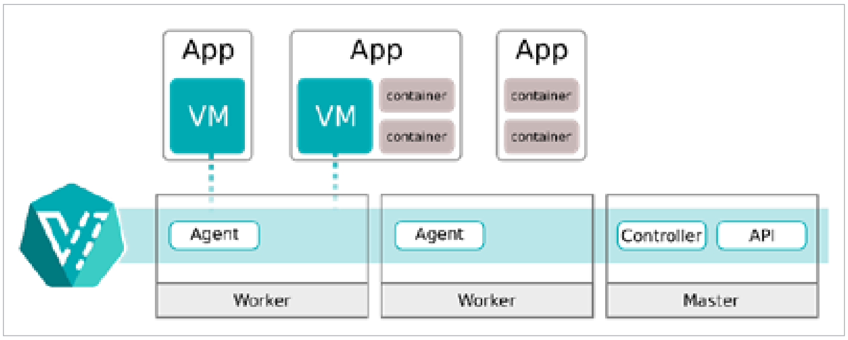

Figure 3: Architecture - KubeVirt user guide

- **Platform9**: Delivers managed Kubernetes as a SaaS solution, offering global cluster management with no infrastructure overhead.
- **Rancher**: Provides open source multi-cluster management, offering automated provisioning, security, and app deployment.

Key FOSS tools include:

- **KubeFed**: Federates clusters, enabling cross-cluster resource management.
- **Submariner**: Enables secure communication between clusters.
- **ArgoCD**: GitOps-based tool supporting automated multi-cluster deployments.

Mastering MCMCK empowers organisations to optimise costs, enhance high availability, and achieve global scalability. By leveraging purpose-built tools like Google Anthos, OpenShift, and open source solutions, enterprises can effectively manage Kubernetes across multi-cloud environments and unlock the full potential of their infrastructure.

Advanced scheduling techniques

Advanced Kubernetes scheduling has significantly evolved, particularly with the rise of serverless computing. Kubernetes v1.30 offers enhanced capabilities for managing and scaling containerised workloads, making it an ideal platform for complex, production-grade applications.

The Kube scheduler, part of Kubernetes' control plane, is responsible for placing newly created or unscheduled pods on suitable nodes. It filters available nodes based on specific criteria and scores them to select the most suitable node for the pod. The scheduler then notifies the API server of the decision through a binding process. Users can also develop custom schedulers tailored to unique workload requirements.

Kubernetes now supports advanced scheduling features to optimise resource allocation for modern, demanding workloads:

Node and pod affinity/anti-affinity: Node affinity places workloads based on specific node characteristics, while pod affinity/anti-affinity manages pod placement relative to each other, improving communication or fault tolerance.

Volcano scheduler: Optimised for HPC, machine learning, and batch processing, Volcano efficiently allocates resources for large-scale computational tasks, improving parallel processing and job dependencies.

KubeVirt: This enables hybrid scheduling of both VMs and containers, simplifying management of diverse workloads on a single Kubernetes platform.

These features enhance Kubernetes' ability to manage both traditional and modern resource-intensive workloads efficiently.